Deep Learning-Based Uplink Transmit Power Estimation in 4G/5G Networks

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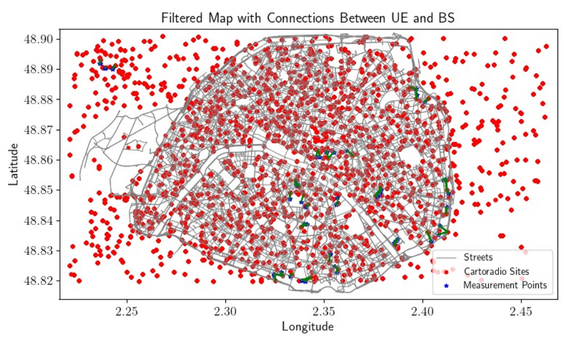
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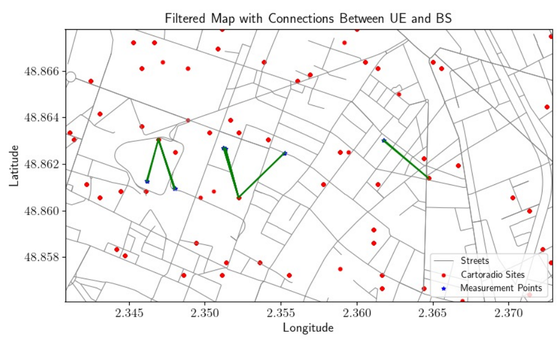
Introduction

In urban environments, regulation on radiofrequency electromagnetic field (RF-EMF) exposure limits the intensity of these fields in wireless communication links between a user (UE) and a base station (BS). Therefore, it is relevant to quantify the power emitted by a user over space and time. However, it is challenging to obtain reliable direct measurements of the E-field intensity, since it requires specialized equipment and time-consuming campaigns. This work proposes to solve this problem by combining heatmap images with a few measurements of downlink (DL) frequencies from BSs to estimate the uplink (UL) power transmitted by a UE in a 4G/5G network at any point, using a deep learning framework.

Data treatment

The data used in this work has two types: the public data, obtained through the Cartoradio website made by ANFR, and the measurements performed by [1]. In the first case, the regions are selected separately and without intersection, and their corresponding datasets are later concatenated into one. Only information about the antenna dimensions, location and system are retained. In the second case, data is collected by testing four different application services at each point. Information about the location, time and energy, frequency, and technology in the wireless link are retained. The criterium for the BS to connect to the UE is being the closest to it while having the same operator, technology and signal frequency.





Deep learning model

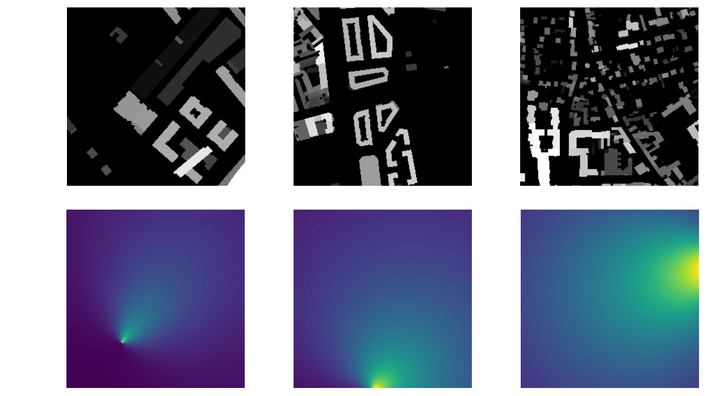
In order to obtain a clear represenation of the space surrounding the UEs and BSs, the coordinates (latitude and longitude) of both are converted from degrees in the world to pixels in an image. This is done by first converting degrees to kilometers through the haversine distance between two points on the surface of the Earth, and then from kilometers to pixels through a constant. All images are squares centered around the user, and they represent either the height of the buildings (height map) or the propagation of the signal (antenna radiation map) that decays logarithmically with the distance:

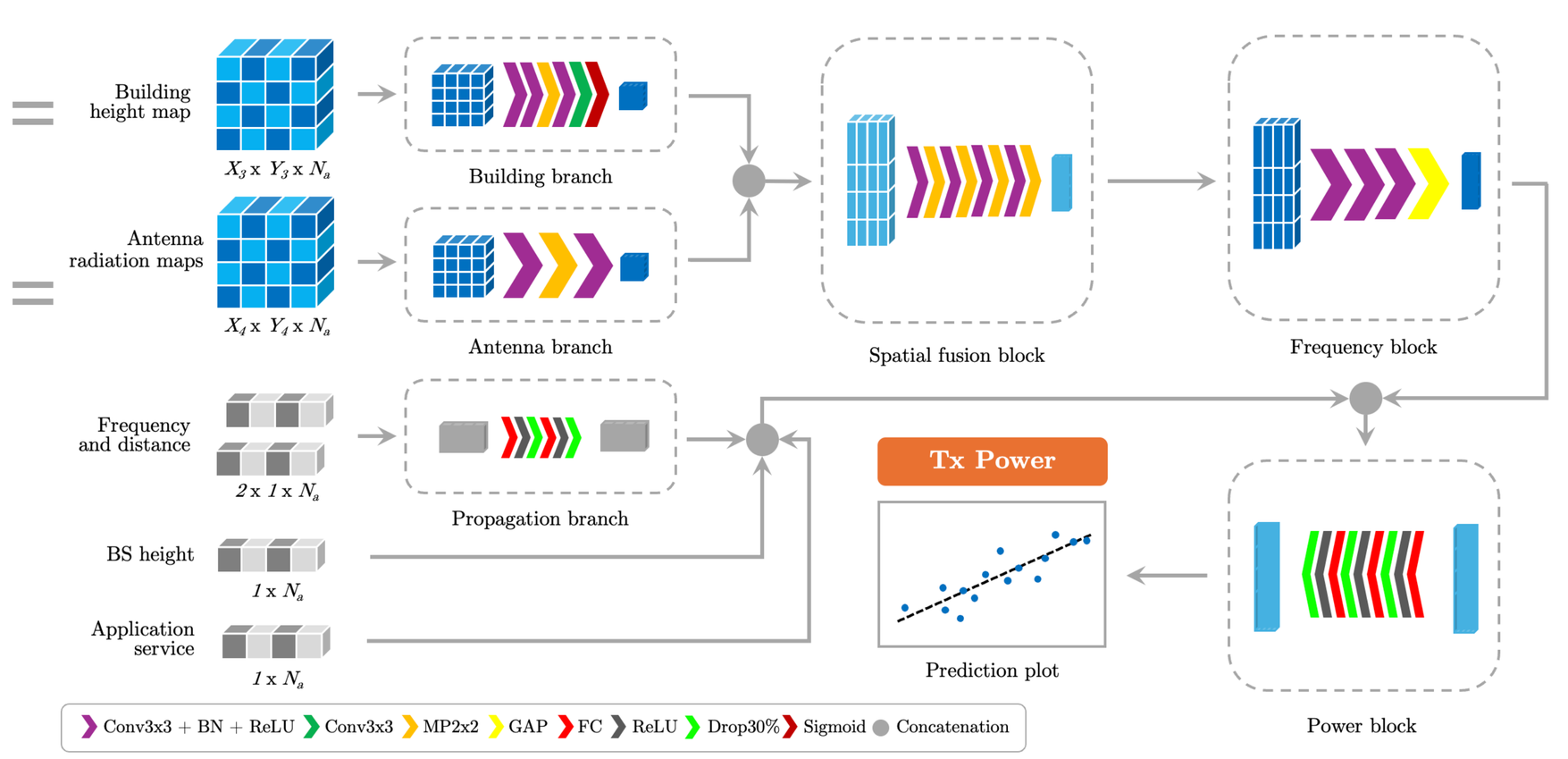
$$FSPL_{dB} = 20\log_{10}(d) + 20\log_{10}(f) + 20\log_{10}\left(\frac{4\pi}{c}\right)$$

They are used as inputs in different branches, but both use a combination of convolutional neural network (CNN) architectures to extract intricate information from figures. The 4G/LTE and 5G technologies are differentiated by their radiation patterns – a cardioid for the former and a normalized von Mises distribution for the latter.

Besides both images, the model takes as inputs the frequency of the signal and the distance [km] between UE and the BS it connects to, the BS height, and one of the four application services (FTP, voice call, video call, VoIP) as a hot-encoded label. They are all column vectors and not images – thus processed by branches with fully connected (FC) layers from the start, which are better strcutured for their shape.

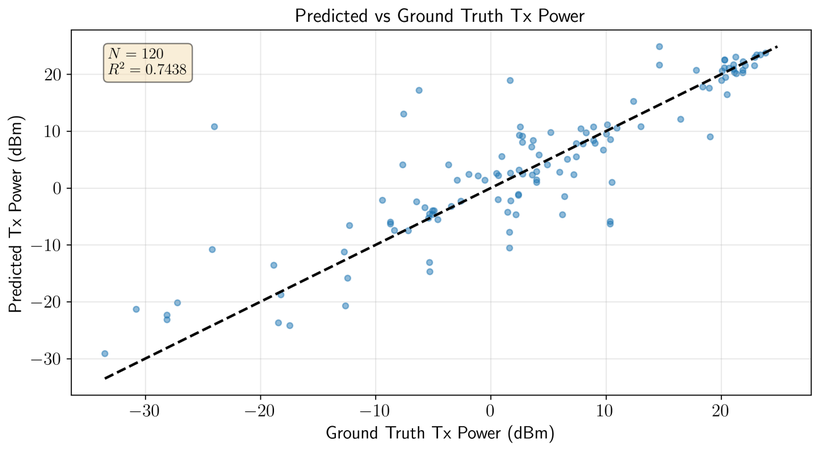
Architecture

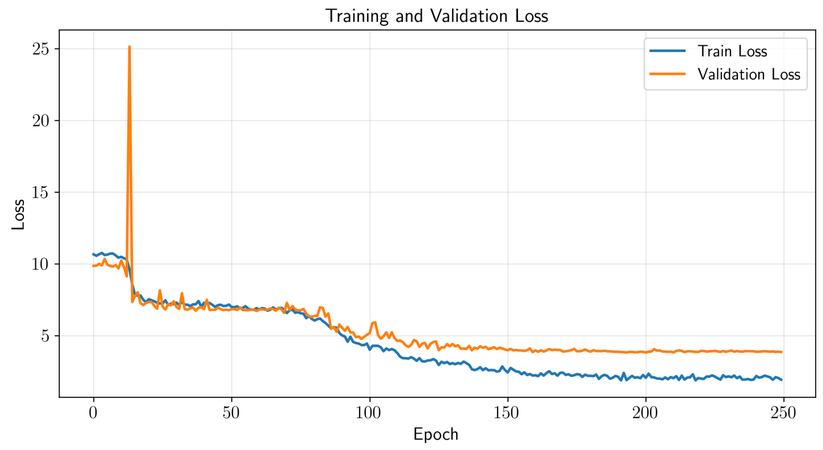




Results

- There are a total of 603 points after filtering, of which 120 (20%) are taken for testing/validation. Their Hubber loss is similar to the training set (other 80%). The R^2 is calculated with respect to the line of perfect relationship (prediction matches the ground truth at all points).
- Higher powers are better predicted than lower powers, since these are more subject to influece of other variables (noise and fading). Still, globally, the score is an improvement with respect to the literature [2] and sufficient for the desired performance, especially due to the limited amount of measure data.





Acknowledgements

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References

[1] J. Liu et al., *Assessment of EMF Exposure Induced by Wireless Cellular Phones in Various Usage Scenarios in France*, IEEE Access, vol. 13, pp. 123743-123755, 2025.

[2] Q. Sun et al., *Measurement-Driven Deep Learning for Uplink RF Exposure Assessment in Urban Environments*, [Preprint].

Code

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